

Winning Space Race with Data Science

SpaceX Falcon 9 First Stage Landing Prediction

Benítez Lorena A.
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EXECUTIVE SUMMARY



Objective

Predict Falcon 9 first-stage landing success using machine learning to help competitors estimate launch costs.



Data Collection

Leveraged SpaceX API and Wikipedia web scraping for technical parameters and historical records.



Methodology

Data cleaning, SQL-based EDA, Python visualizations, and interactive mapping with Folium.



Results

Logistic Regression, SVM, and KNN tied with a peak accuracy of 83.33%, while the Decision Tree performed worst at 66.67%.

INTRODUCTION:



Aerospace Disruption

The Reusability Milestone: SpaceX revolutionized the industry by successfully landing and reusing the Falcon 9 first-stage booster, dramatically decreasing orbit access costs.

Commercial Advantage: Lowered launch prices to approximately **\$62M**, compared to traditional competitors whose costs exceed **\$165M**.



Problem Statement

Predictive Value: Understanding landing variables allows competitors to accurately estimate launch costs and predict market pricing.

Data-Driven Approach: Leveraging data engineering, geospatial mapping, and machine learning to predict outcomes based on historical flight data.

DATA METHODOLOGY

COLLECTION & WRANGLING



METHODOLOGY



Data Collection

Falcon 9 data via SpaceX API and Wikipedia scraping (BeautifulSoup).



Data Wrangling

Handling missing values and engineering the binary Class target variable.



EDA (SQL & Visualization)

SQL metrics extraction and visual trend analysis via Matplotlib/Seaborn.



Interactive Analytics

Folium site mapping and Plotly Dash apps for dynamic payload impact analysis.



Predictive Analysis

Deployed LogReg, SVM, Decision Tree, and KNN classification models.



Evaluation & Results

Hyperparameter tuning via GridSearchCV achieved 83.33% model accuracy.

DATA COLLECTION

Methodology Overview

API Integration

RESTful API using Requests library for dynamic data access.

Web Scraping

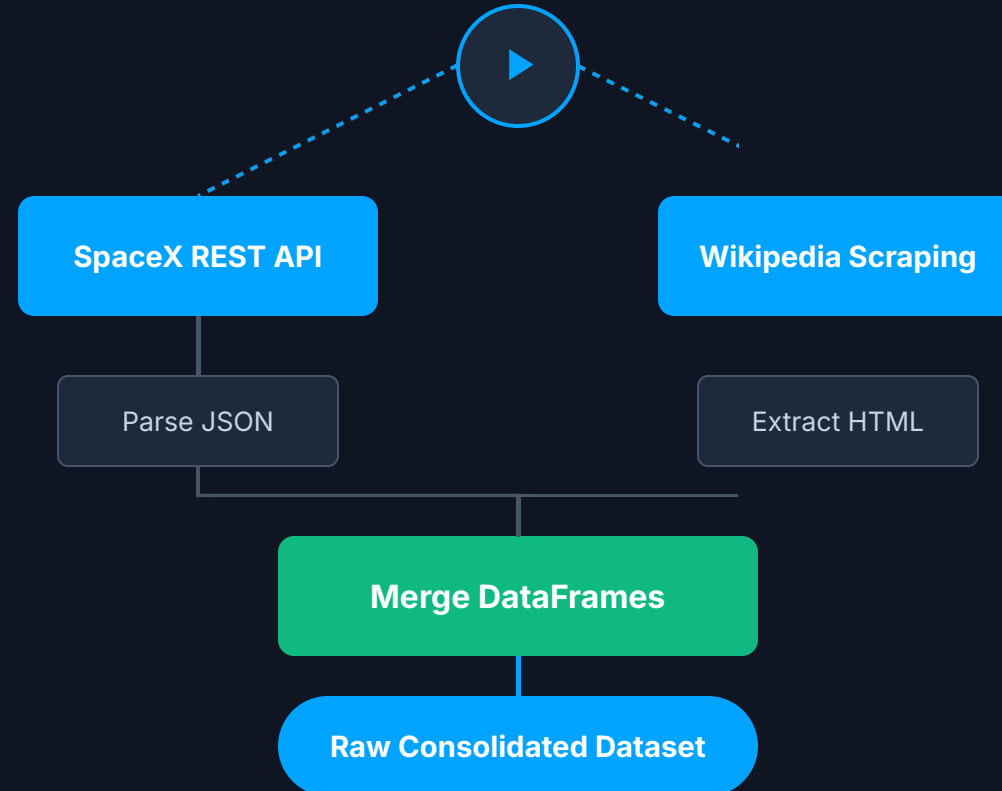
BeautifulSoup4 for efficient HTML parsing of unstructured web data.

Data Formats

Converting raw JSON and HTML tables into structured Pandas DataFrames.

Consolidation

Merging disparate sources into a single, analysis-ready dataset.



DATA COLLECTION:

➤ SPACEX API

Integration Details

API Endpoint

Targeted SpaceX v4/launches/past for historical mission records.

HTTP Request

Python requests library utilized for efficient RESTful GET execution.

Data Parsing

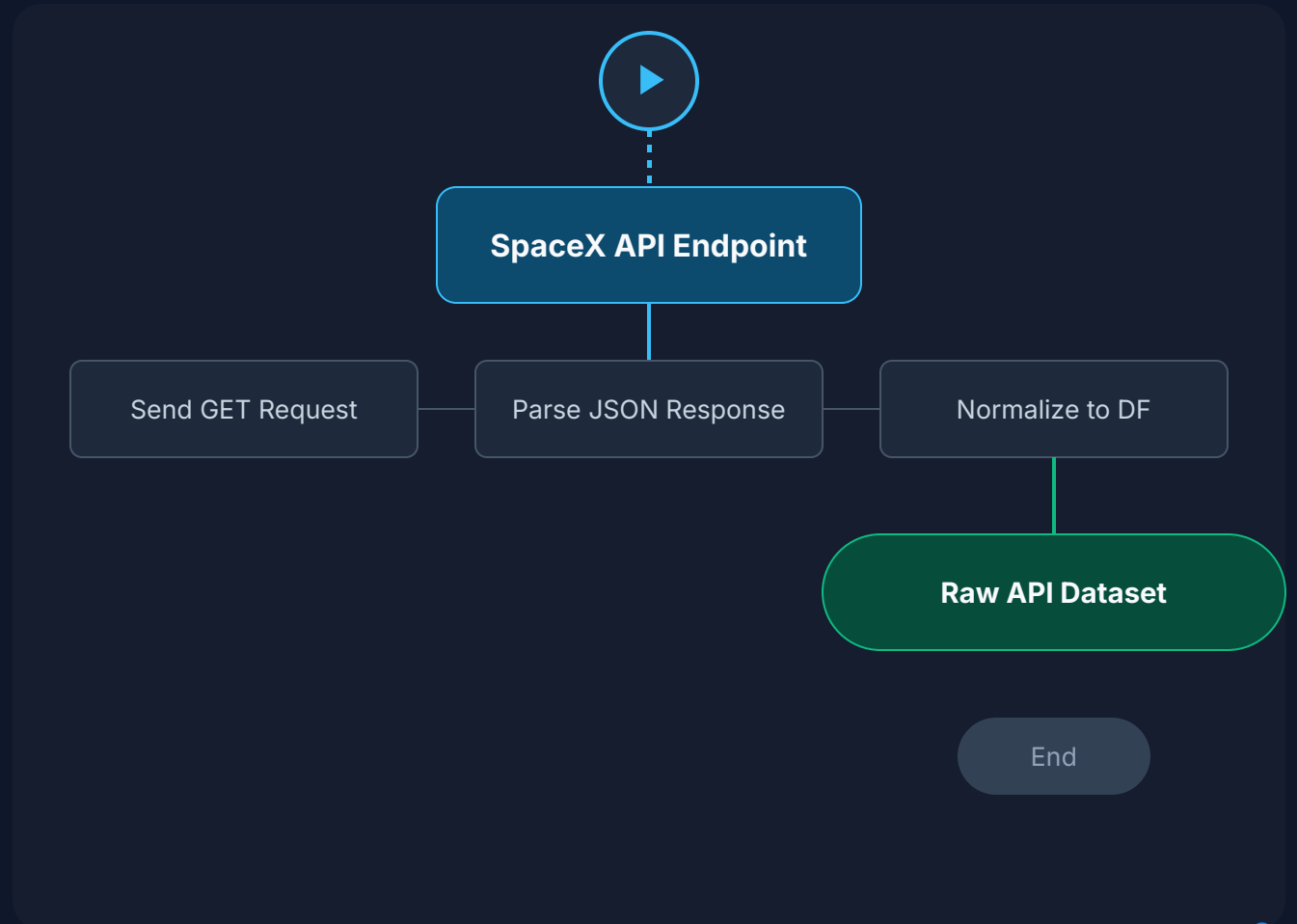
Extracted payload from JSON response into structured dictionaries.

Data Structuring

Flattened nested objects into Pandas DataFrames via `json_normalize()`.

GitHub Reference:

<https://github.com/NiNaCocaZero/SpaceX-Falcon9-Predictive-Analysis>



DATA COLLECTION:

➤ WEB SCRAPING

Extraction Methodology

Target URL

Wikipedia page for Falcon 9 and Falcon Heavy launches.

HTML Request

Raw HTML content retrieved using the `requests` library.

HTML Parsing

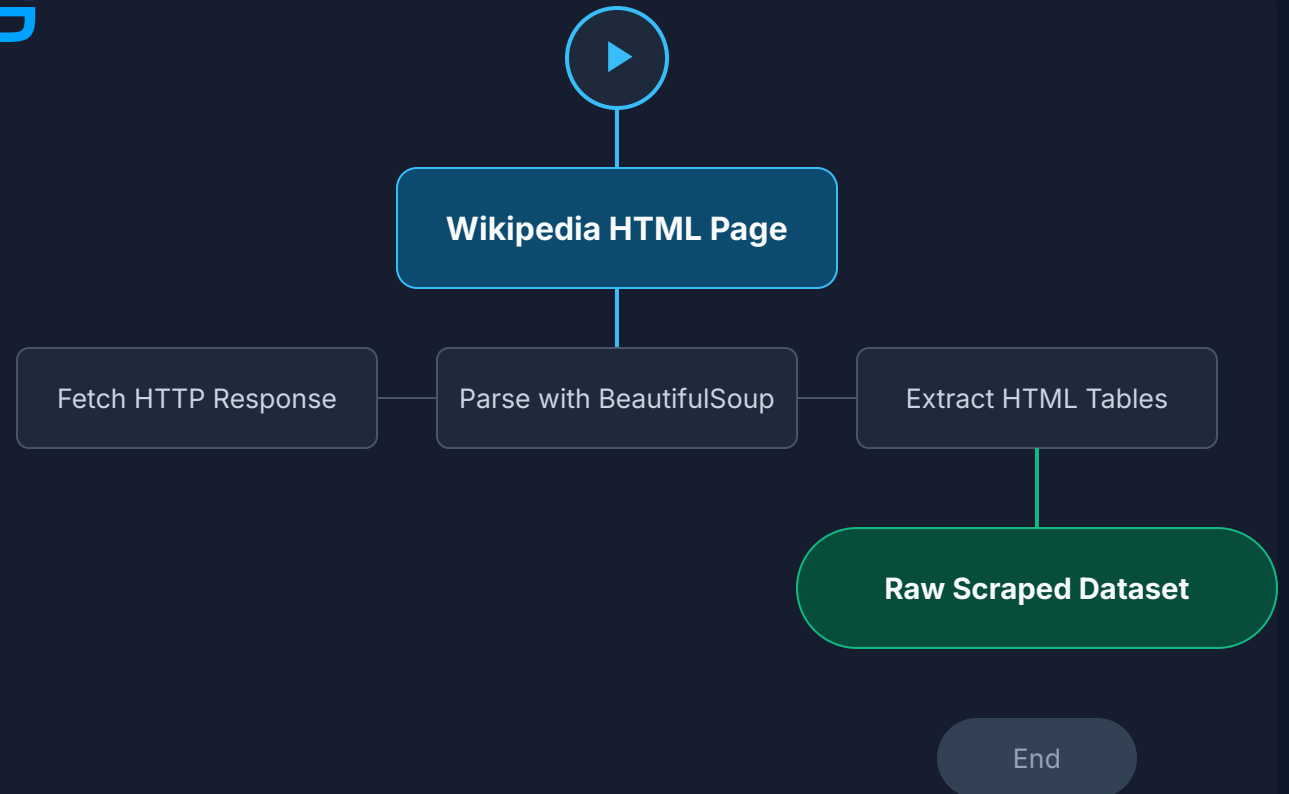
`BeautifulSoup` utilized to navigate and locate data tables.

Data Extraction

Iterated through `<tr>` and `<th>` tags to populate DataFrames.

GitHub Reference:

<https://github.com/NiNaCocaZero/SpaceX-Falcon9-Predictive-Analysis>



EDA WITH DATA VISUALIZATION

Scatter Plots

Correlation Analysis

Used to identify correlations between continuous variables (Payload Mass & Flight Number vs. Outcome).

Key Insight

Helped visualize if heavier payloads or experienced boosters directly influenced first-stage landing success.

Bar Charts

Categorical Comparison

Ideal for comparing historical success rates across different target orbits (LEO, ISS, GEO).

Strategic Value

Determined which missions are inherently riskier based on orbital requirements.

Line Charts

Time-Series Analysis

The industry standard for tracking the evolution of reliability over long periods.

Reliability Trend

Confirmed the consistent upward trend in SpaceX's landing success rates year over year.

EDA with Data Visualization

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EDA WITH SQL

Total Payload Mass

Calculated the aggregate payload mass carried by the Falcon 9 boosters across all missions to understand lifting capacity.

Orbit Averages

Queried the average payload mass specifically for the ISS (International Space Station) orbit to identify standard mission requirements.

Launch Site Success

Filtered records to determine the launch sites with the highest success rates, highlighting **KSC LC-39A** as a top performer.

Historical Milestones

Extracted the date of the first successful landing outcome on a ground pad using the **MIN()** function on the date column.

GitHub Reference:

https://github.com/NiNaCocaZero/SpaceX-Falcon9-Predictive-Analysis/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

INTERACTIVE GEOSPATIAL MAPPING

Strategic Insights

Geospatial Visualization

Used **Folium** to plot launch sites and visualize proximity to critical infrastructure.

Marker Clusters

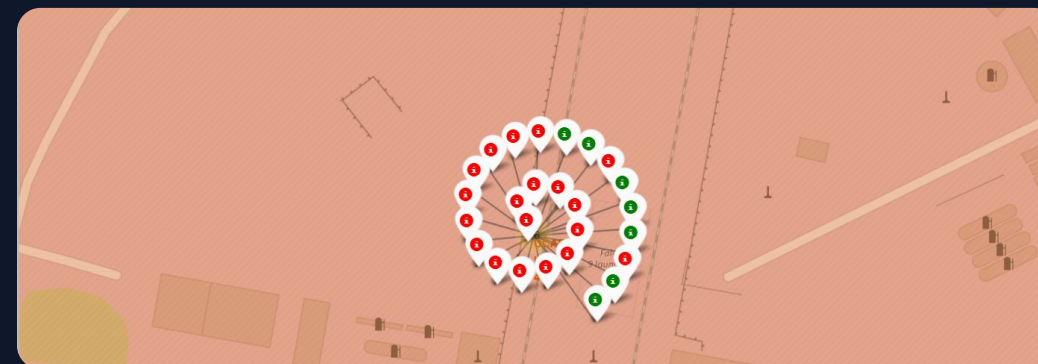
Implemented clusters to manage high-density data points and improve map legibility.

Proximity Analysis

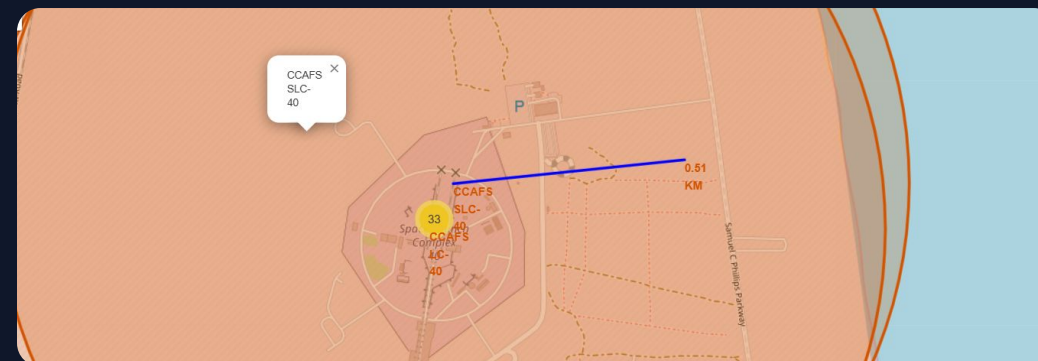
Calculated distances to coastlines, railways, and highways via the Haversine formula.

Safety & Logistics

Confirmed sites are strategically near coasts for safety and rail for booster transport.



Launch outcomes clustered by site (Green: Success, Red: Failure).



Distance calculation (0.51 km) from the launch pad to the closest coastline.

GitHub Reference:

[https://github.com/NiNaCocaZero/SpaceX-Falcon9-Predictive-Analysis/blob/main/lab_ipynb_launch_site_location%20\(2\).ipynb](https://github.com/NiNaCocaZero/SpaceX-Falcon9-Predictive-Analysis/blob/main/lab_ipynb_launch_site_location%20(2).ipynb)

BUILD A DASHBOARD

➤ PLOTLY DASH

Dashboard Features

Interactive Exploration

Developed a user-friendly web dashboard using **Plotly Dash** to filter and visualize historical launch data in real-time.

Site Success Distribution

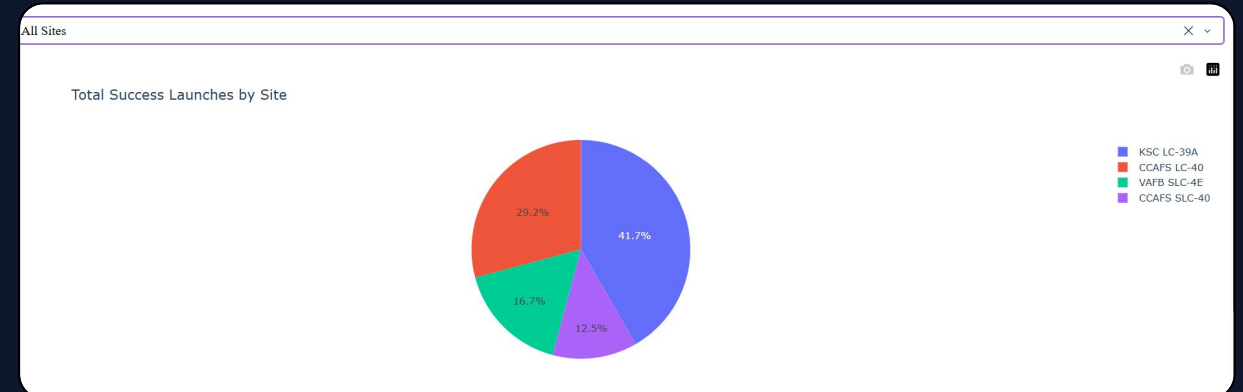
Implemented dynamic pie charts to analyze the proportion of successful versus failed landings for each specific launch site.

Payload vs. Outcome

Created interactive scatter plots to correlate payload mass (kg) with landing success, utilizing color-coding for Booster Versions.

GitHub Reference:

<https://github.com/NiNaCocaZero/SpaceX-Falcon9-Predictive-Analysis/blob/main/spacex-dash-app.py>



Success rate distribution for the selected launch site.



Correlation between payload mass and landing outcome, categorized by Booster Version.

PREDICTIVE ANALYSIS FRAMEWORK

End-to-End Classification Pipeline



Data Preprocessing

Standardized features using `StandardScaler` to ensure uniform scale across the dataset.



Validation Strategy

Performed an 80/20 `train_test_split` to ensure robust validation and prevent overfitting.



Model Optimization

Trained LR, SVM, Tree, and KNN. Applied `GridSearchCV` (10-fold CV) for optimal tuning.



Performance Metrics

Evaluated using Accuracy Scores and Confusion Matrices to identify the champion model.

GitHub Reference:

https://github.com/NiNaCocaZero/SpaceX-Falcon9-Predictive-Analysis/blob/main/SpaceX_Machine_Learning_Prediction_Part_5.ipynb

MACHINE LEARNING RESULTS

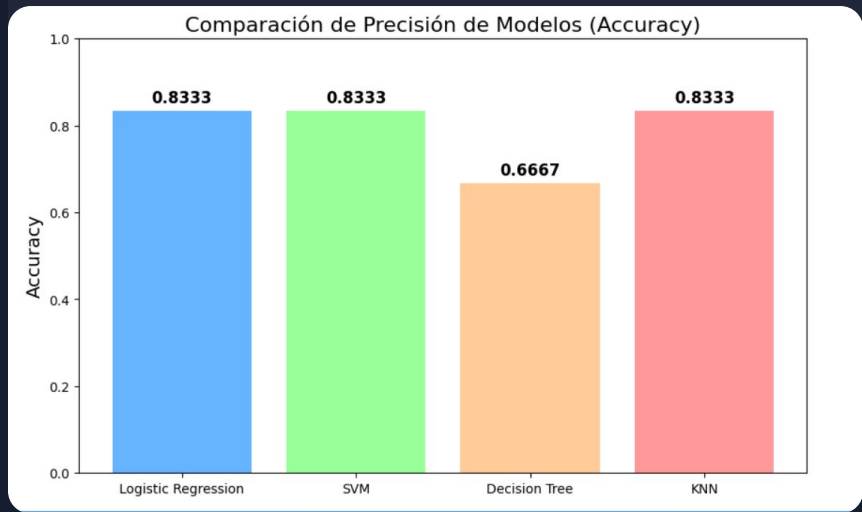
Key Findings

Accuracy

Logistic Regression, SVM, and KNN achieved the highest test accuracy (83.33%), while Decision Tree obtained a lower score (66.67%).

Hyperparameter Convergence

Extensive tuning via **GridSearchCV** resulted in synchronized success across models.



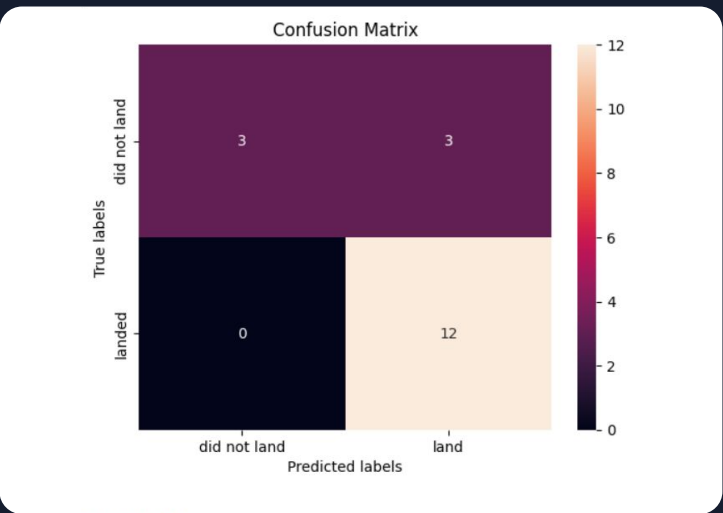
Accuracy score comparison across tested classification algorithms.

Reliable Predictions

Confusion matrices confirm high reliability with minimal false positives for rocket landing success.

Selection for Deployment

Logistic Regression or SVM are preferred due to mathematical simplicity and efficiency.



Confusion matrix demonstrating high predictive precision for landings.

EXPLORATORY ANALYSIS

Insights via SQL & Python



FLIGHT NUMBER VS. LAUNCH SITE



Operational Evolution

Positive Trajectory

The line chart demonstrates a clear upward trajectory in launch operations and success.

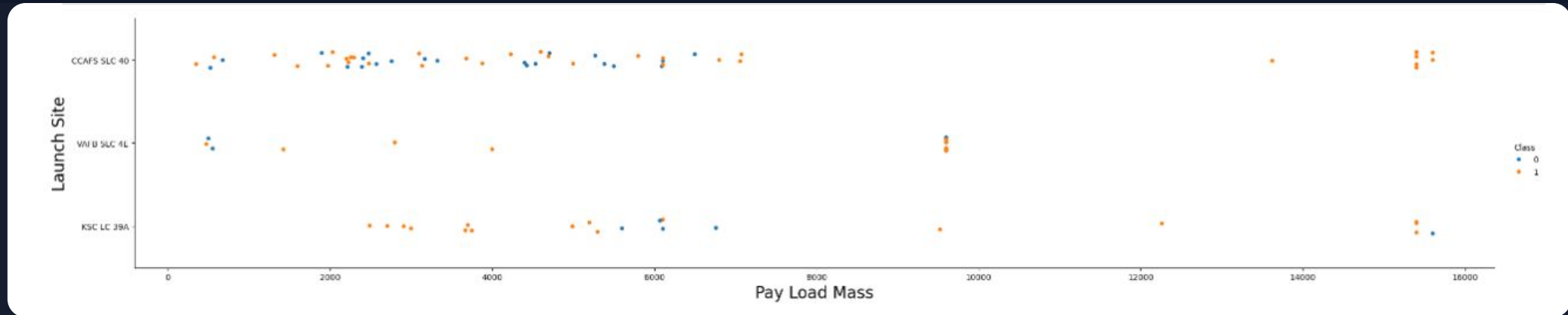
Process Maturity

From a turbulent start in 2013, SpaceX dramatically improved its first-stage recovery process.

Modern Excellence

SpaceX reached **near-perfect success rates** by 2020 through iterative engineering.

PAYLOAD VS. LAUNCH SITE



Scatter plot illustrating the relationship between Payload Mass (kg) and Launch Site success.

Key Capacity Insights



Heavy Payload Dominance

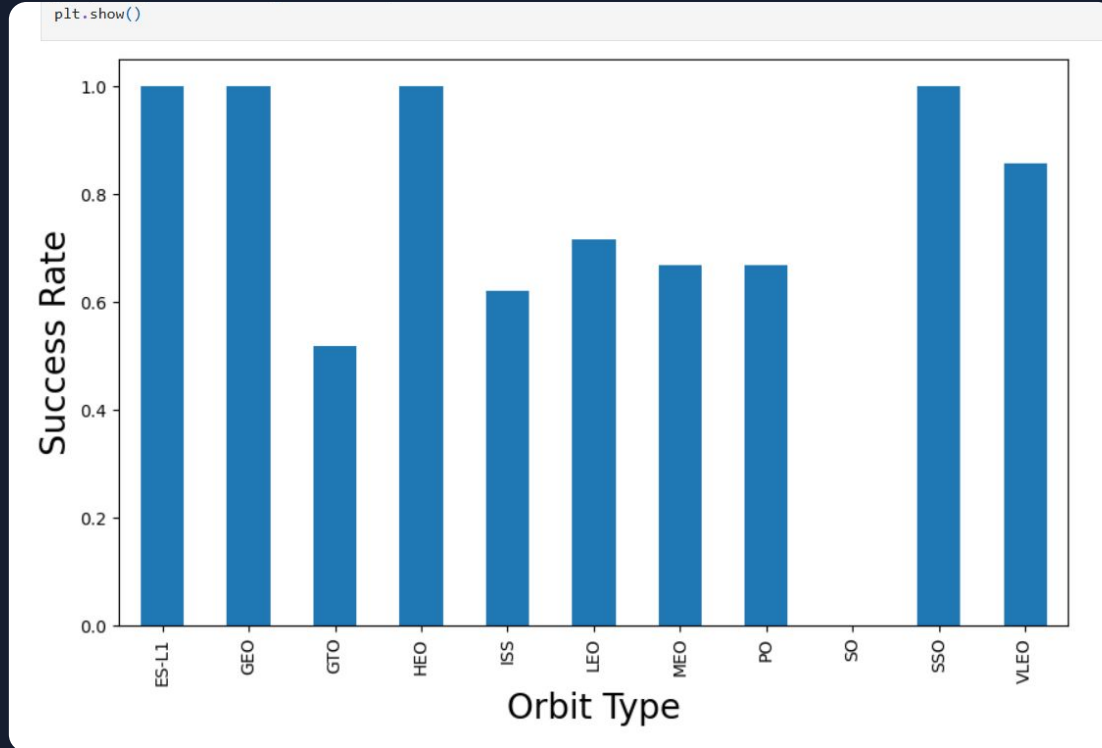
VAFB SLC-4E and KSC LC-39A tend to handle heavier payloads, reaching up to 10,000 kg.



Operational Consistency

CCAFS SLC-40 launches a wider variety of masses; success remains highly consistent for payloads under 6,000 kg.

SUCCESS RATE VS. ORBIT TYPE



Success rate distribution across target orbits.

Orbit Reliability Analysis

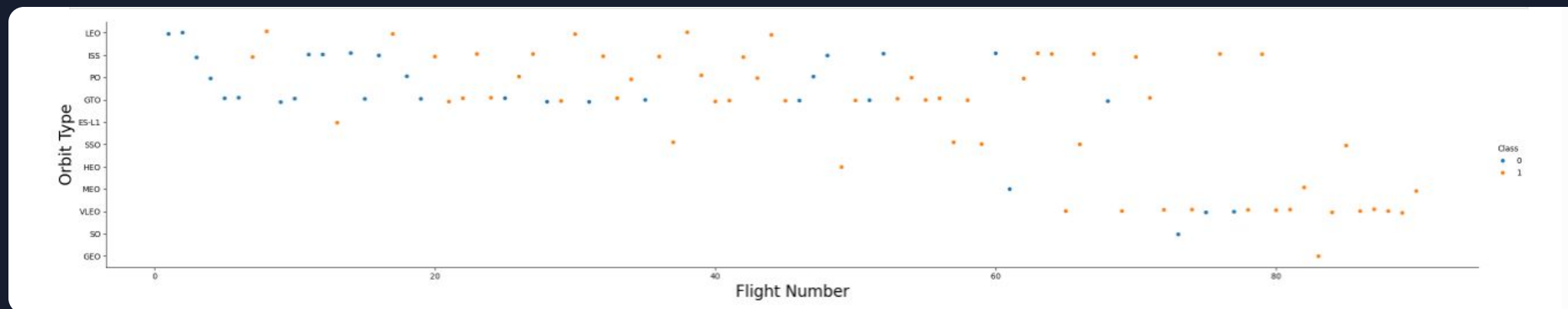
Peak Performance

Orbits like ES-L1, GEO, HEO, and SSO have achieved a **perfect 100% success rate**.

Technical Challenges

GTO (Geosynchronous Transfer Orbit) exhibits a lower success rate, reflecting the increased complexity and fuel requirements for these missions.

FLIGHT NUMBER VS. ORBIT TYPE



Scatter plot displaying the relationship between Flight Number and target Orbit Type, colored by landing class.

Strategic Orbit Mastery

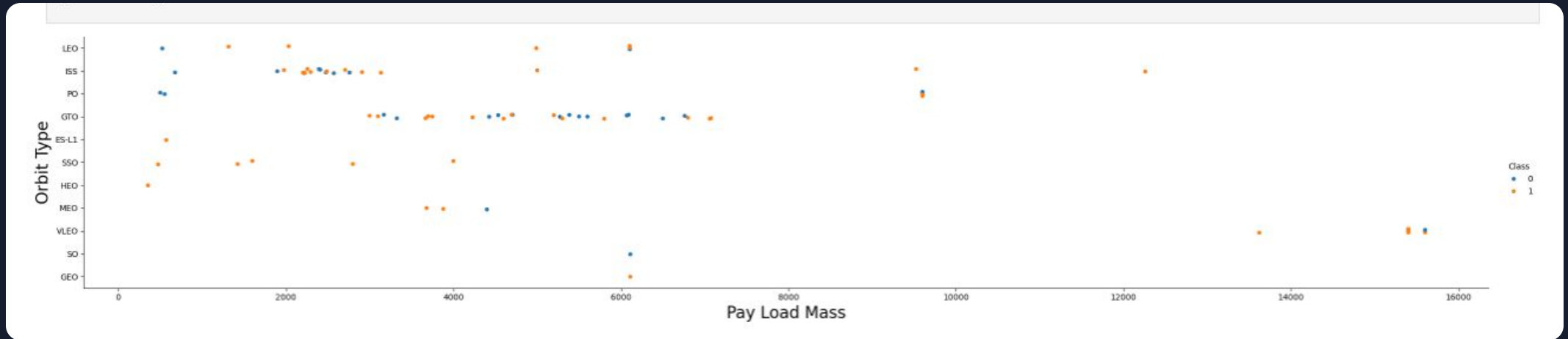
Concentrated Mission Density

- As flight numbers increase, missions become heavily concentrated in specific orbits like VLEO and ISS.

Recovery Process Optimization

- Landing success is highly consistent in these frequent orbits, demonstrating SpaceX mastered the recovery process for common commercial routes over time.

PAYLOAD VS. ORBIT TYPE



Scatter plot illustrating the distribution of Payload Mass (kg) across target Orbits, categorized by landing outcome.

Payload Capability Analysis

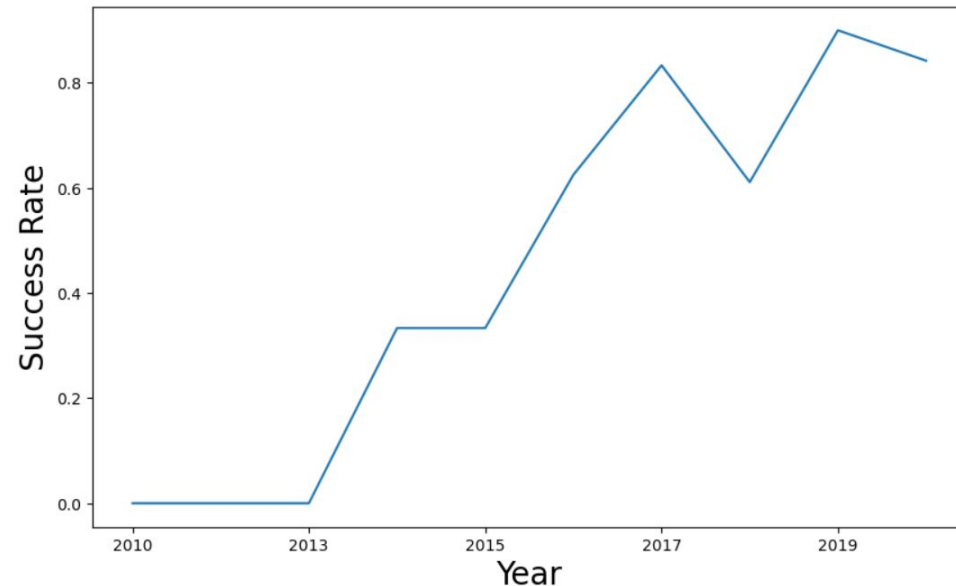
Strategic Mission Distribution

- Heavier payloads (> 6,000 kg) are predominantly directed towards Polar Orbit (PO) and the International Space Station (ISS).

Landing Performance Resilience

- The data suggests that Falcon 9 maintains high landing success rates even for heavier payloads.

LAUNCH SUCCESS YEARLY TREND



Line chart tracking the yearly average success rate of Falcon 9 first-stage landings.

Mission Mastery Evolution

Exponential Growth Trend

- The data reveals a strong upward trend in landing success, starting from 0% in 2013 and climbing steadily to nearly 100% by 2020.

Engineering Excellence

- This progression clearly illustrates SpaceX's rapid technological advancement, overcoming the initial learning curve to achieve mastery in reusable rocket engineering over the decade.

ALL LAUNCH SITE NAMES

```
In [13]: %%sql
        SELECT DISTINCT "Launch_Site"
        FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[13]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

SQL query output displaying the distinct launch sites identified within the SpaceX mission dataset.

Database Query Insights

Unique Operational Bases

- The SQL query utilizes the **DISTINCT** keyword to filter the dataset and isolate the unique operational bases used by SpaceX.

Strategic Footprint

- The results reveal primary facilities like CCAFS, VAFB, and KSC, highlighting a strategic geographical footprint for diverse mission profiles.

LAUNCH SITE NAMES: 'CCA' PATTERN

```
%sql
SELECT * FROM SPACEXTABLE
WHERE "Launch_Site" LIKE 'CCA%'
LIMIT 5

* sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Database Query Insights

Pattern Matching Efficiency

➤ The SQL query employs the **LIKE** operator and **%** wildcard to isolate sites starting with 'CCA'.

Geographical Specificity

➤ Results successfully isolate Cape Canaveral (CCAFS) facilities, verifying the regional scope of the historical launch data.

SQL query results filtering for Cape Canaveral Air Force Station (CCAFS) launch sites.

TOTAL PAYLOAD MASS FOR NASA

```
In [19]: %%sql
SELECT SUM(PAYLOAD_MASS_KG_)
FROM SPACEXTABLE
WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[19]: SUM(PAYLOAD_MASS_KG_)
          45596
```

Database Query Insights

Aggregate Performance

- The SQL query applies the `SUM()` aggregate function to the `PAYLOAD_MASS_KG_` column, utilizing a `WHERE` clause to filter specifically for NASA missions.

Mission Scale

- This calculation reveals the total accumulated mass SpaceX has launched for NASA, demonstrating the substantial lifting capacity of the Falcon 9 and the massive scale of their partnership.

SQL query output aggregating the total payload weight for all historical NASA (CRS) missions.

AVERAGE PAYLOAD MASS BY F9 V1.1

```
3]: %%sql
SELECT AVG("PAYLOAD_MASS_KG_")
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1'

* sqlite:///my_data1.db
Done.
3]: AVG("PAYLOAD_MASS_KG_")
2928.4
```

SQL query result determining the average payload mass for the F9 v1.1 variant.

Database Query Insights

Statistical Precision

- The SQL query utilizes the `AVG()` function to determine the mean payload mass specifically for the F9 v1.1 booster variant.

Technical Baseline

- This calculation establishes a technical baseline for the typical mission requirements and lifting capacity handled by these earlier versions of the Falcon 9 fleet.

FIRST SUCCESSFUL GROUND LANDING DATE

```
In [29]: %%sql
          SELECT MIN("Date")
          FROM SPACEXTABLE
          WHERE "Landing_Outcome" = 'Success (ground pad)';

* sqlite:///my_data1.db
Done.

Out[29]: MIN("Date")
         2015-12-22
```

SQL query identifying the first successful Falcon 9 landing on a solid ground pad.

Database Query Insights

Historical Milestone

- The SQL query identifies the exact historical milestone by utilizing the `MIN()` function on the date column, specifically filtering for 'Success (ground pad)' outcomes.

Pivotal Moment

- This record marks the first time SpaceX successfully returned a Falcon 9 first stage to solid ground, a key step in rocket reusability.

SUCCESSFUL DRONE SHIP LANDING: 4K-6K PAYLOAD

List the names of the boosters which have success in drone

%%sql

```
SELECT "Booster_Version"  
FROM SPACEXTABLE  
WHERE "Landing_Outcome" = 'Success (drone ship)'  
AND "PAYLOAD_MASS__KG_" BETWEEN 4000 AND 6000;
```

* sqlite:///my_data1.db

Done.

Booster_Version

SQL query results for successful landings with intermediate-weight payloads.

Database Query Insights

Multi-Filter Precision

- The SQL query effectively combines multiple filters: identifying successful drone ship landings while restricting payload mass to a specific range (4,000 to 6,000 kg).

Booster Versatility

- Results highlight specific boosters like F9 FT B1022 and B1026, showcasing Falcon 9's ability to recover stages during demanding missions.

Total Number of Successful and Failed Mission Outcomes

```
%%sql
```

```
SELECT "Mission_Outcome", COUNT(*)  
FROM SPACEXTABLE  
GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db
```

Done.

Mission_Outcome	COUNT(*)
-----------------	----------

Success	101
---------	-----

SQL query results categorizing historical flight outcomes by mission success.

Database Query Insights

Outcome Categorization

- The SQL query utilizes the `COUNT()` aggregate function paired with a `GROUP BY Mission_Outcome` clause to categorize and tally the final results of all historical flights.

Reliability Overview

- The output clearly breaks down the exact number of fully successful missions versus those that experienced anomalies, providing a definitive, high-level overview of SpaceX's operational reliability.

BOOSTERS THAT CARRIED MAXIMUM PAYLOAD

```
In [34]: %%sql
        SELECT "Booster_Version"
        FROM SPACEXTABLE
        WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_")FROM SPACEXTABLE);

* sqlite:///my_data1.db
Done.
Out[34]:
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

SQL subquery results identifying boosters with the absolute maximum payload mass.

Database Query Insights

Maximum Weight Identification

- The SQL query employs a subquery to first determine the absolute maximum payload mass recorded in the dataset, and then filters the main table to identify the specific boosters that carried this exact weight.

F9 Block 5 Superiority

- The results highlight the F9 B5 series (e.g., B1048, B1049, B1051), confirming that this advanced booster version was specifically engineered to handle SpaceX's heaviest deployments, routinely carrying 15,600 kg to orbit.

2015 LAUNCH RECORDS

```
%%sql
SELECT SUBSTR("Date", 6, 2) AS "Month", "Landing_Outcome", "Booster_Version", "Launch_Site"
FROM SPACEXTABLE
WHERE "Date" LIKE '2015%'
AND "Landing_Outcome" = 'Failure (drone ship)';
```

* sqlite:///my_data1.db

Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

SQL query results isolating 2015 'Failure (drone ship)' mission outcomes.

Database Query Insights

Iterative Development Phase

- The SQL query filters historical data to isolate the 2015 timeframe, specifically extracting 'Failure (drone ship)' outcomes alongside booster versions and launch sites.

Path to Recovery Mastery

- These records document the critical iterative testing stages that occurred just before SpaceX successfully mastered ocean-based booster recovery.

RANKED LANDING OUTCOMES (2010-2017)

```
In [36]: %%sql
SELECT "Landing_Outcome", COUNT(*) AS "Cantidad"
FROM SPACEXTABLE
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY "Cantidad" DESC;

* sqlite:///my_data1.db
Done.

Out[36]:
```

Landing_Outcome	Cantidad
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

SQL query results ranking landing frequency during SpaceX's early developmental era.

Database Query Insights

Historical Performance Analysis

- The SQL query filters landing outcomes within a historically significant developmental window (2010 to early 2017) and ranks them in descending order based on their frequency.

Engineering Learning Curve

- The output reveals that "No attempt" and various failure modes were the most common outcomes, illustrating the steep learning curve faced before standardizing recovery procedures.

GEOSPATIAL INTELLIGENCE

Mapping Launch Proximities



GEOGRAPHICAL DISTRIBUTION OF SPACEX LAUNCH SITES

Geographic Strategy

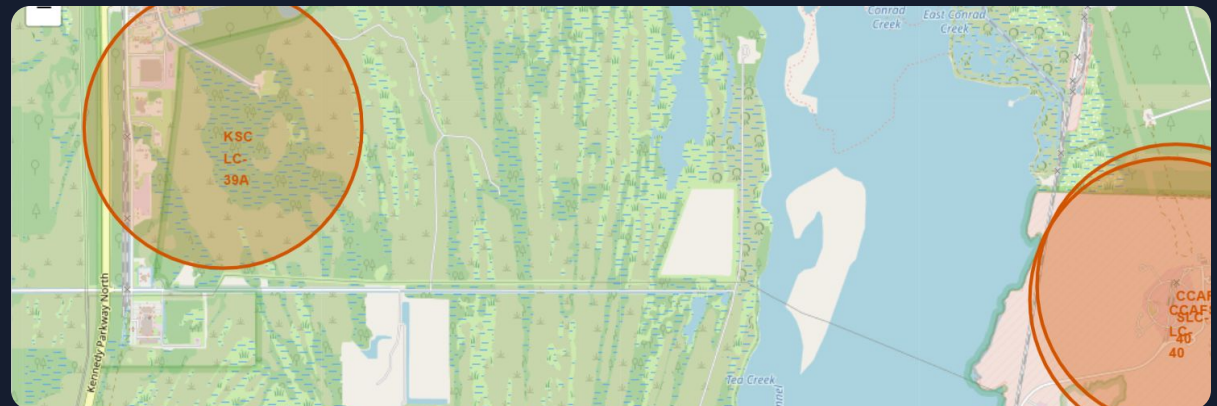
All launch sites are strategically positioned along coastal regions to ensure that rocket debris and first-stage landing attempts occur over open water, maximizing safety for populated areas.

Rotational Boost

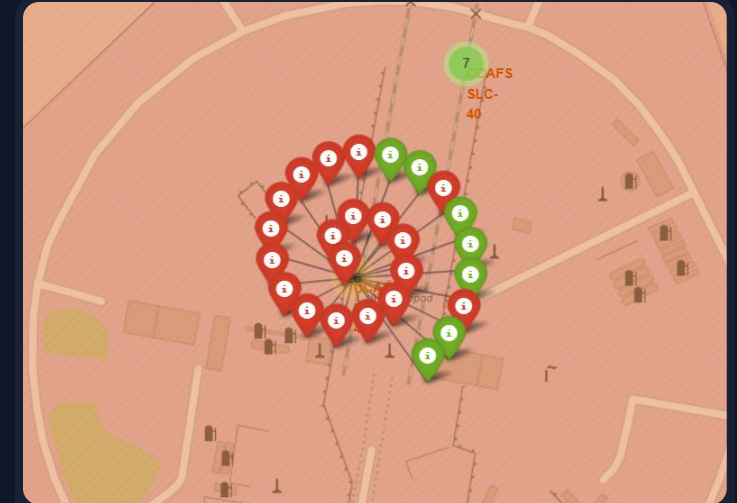
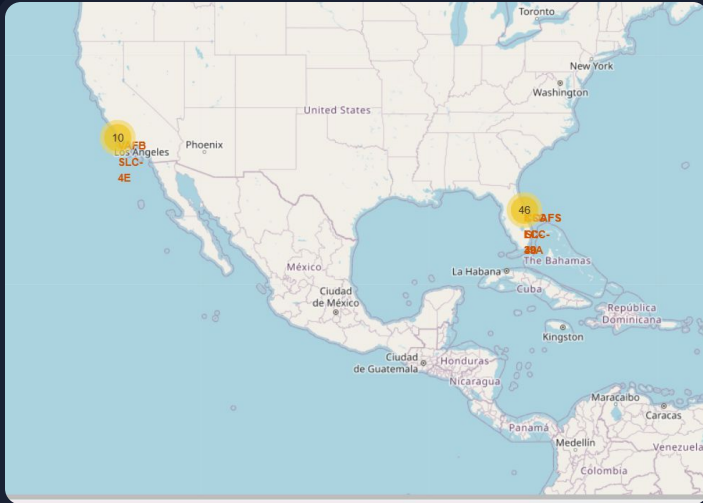
Sites in Florida (CCAFS/KSC) are located at lower latitudes to leverage the Earth's rotational velocity, providing a natural efficiency boost for equatorial orbit insertions.

Operational Diversity

The sites are distributed between the East and West coasts to handle a wide range of mission profiles, including ISS resupply and polar orbit deployments.



LAUNCH OUTCOME ANALYSIS VIA FOLIUM MAPPING



Marker Clustering Strategy: The Folium map utilizes a Marker Cluster plugin to visually categorize launch outcomes, with **green markers** indicating successful first-stage landings and **red markers** indicating failures.

Operational Hub Maturity: This spatial distribution highlights that **KSC LC-39A** and **CCAFS SLC-40** have the highest concentration of successful recoveries, reflecting their maturity as primary operational hubs.

PROXIMITY ANALYSIS & COASTAL STRATEGY

Infrastructure Proximity

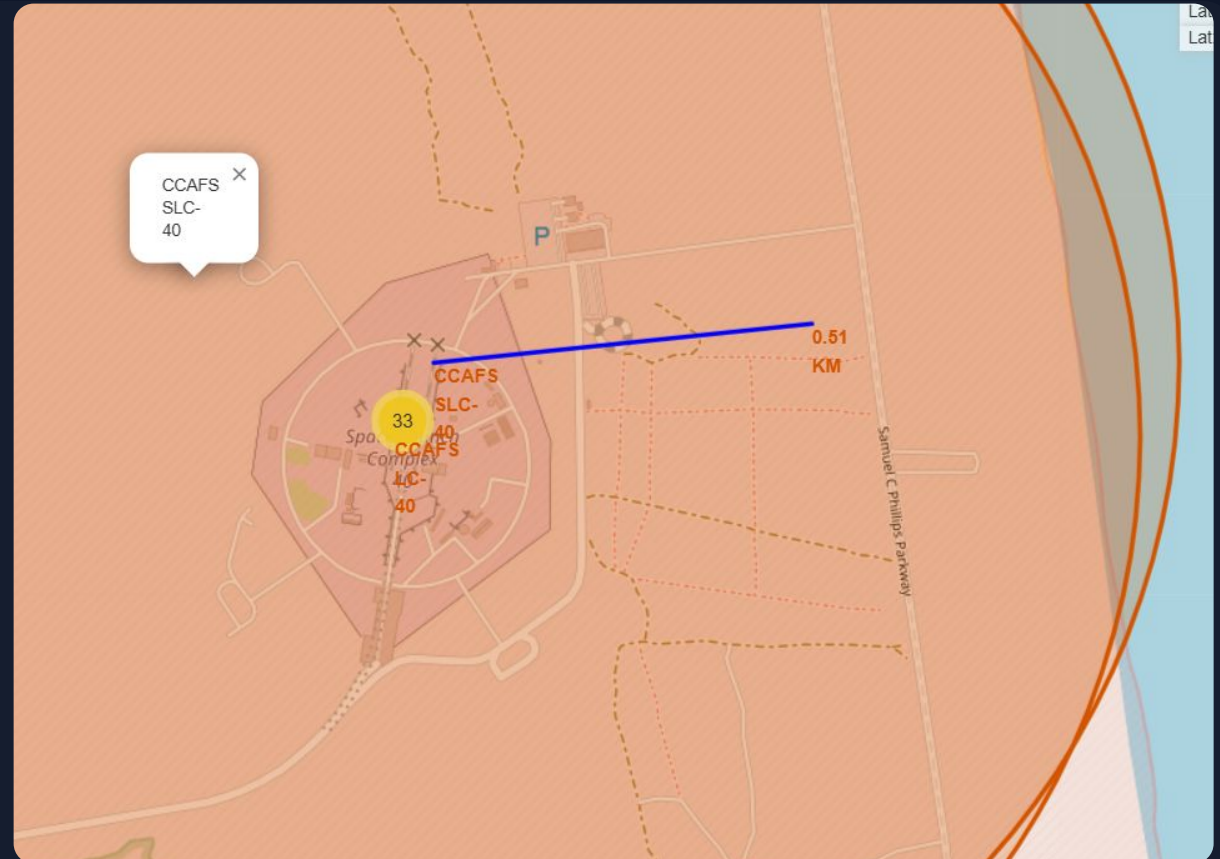
The proximity analysis calculates the direct distance from the launch pad to critical surrounding infrastructure, focusing on the nearest coastline.

Strategic Positioning

The map demonstrates that sites are deliberately positioned extremely close to the ocean (e.g., within a few kilometers) to minimize risk to populated areas.

Booster Recovery

Proximity to the coast facilitates rapid drone ship positioning for efficient booster recovery operations.



MACHINE LEARNING

Predictive Modeling Results



Total Launch Success Overview by Site

Interactive Dashboard

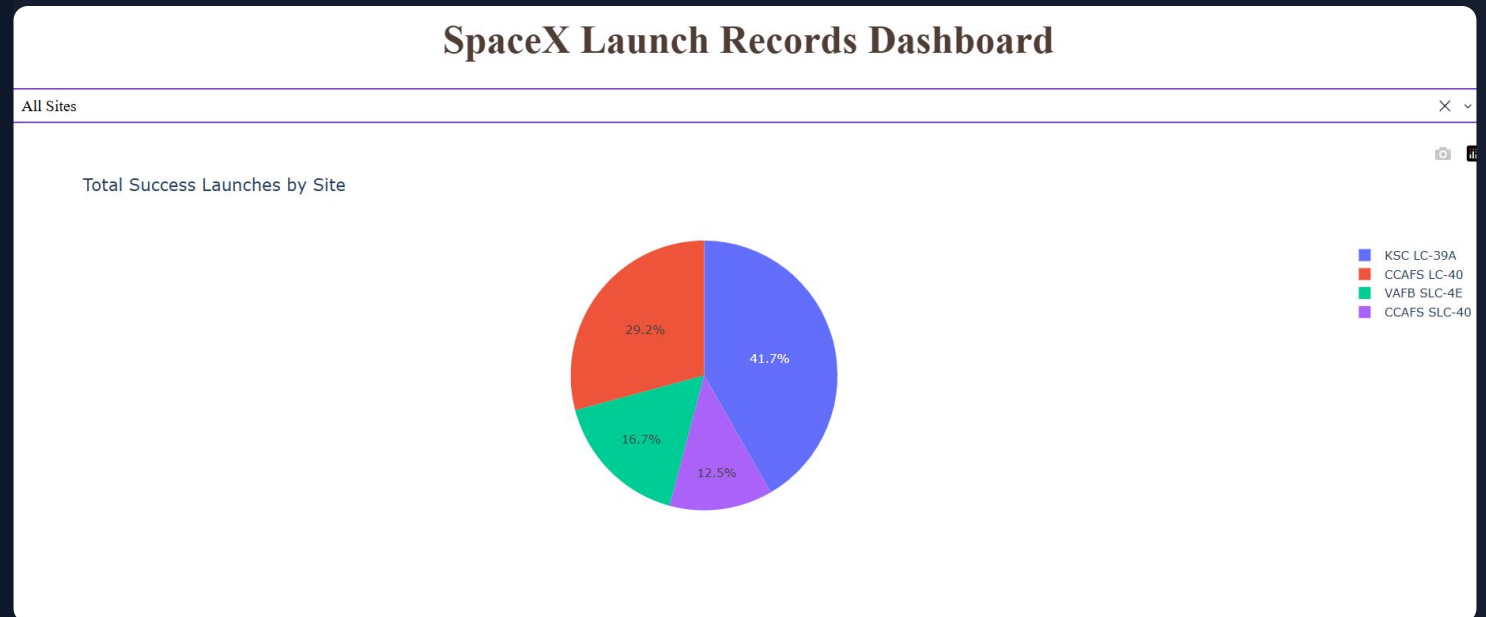
The Plotly dashboard features a pie chart displaying the proportional contribution of each launch site to total successful missions using the "All Sites" filter.

Top Performing Facility

KSC LC-39A emerges as the most prolific facility, accounting for nearly 42% of total successful missions in the dataset.

Strategic Insights

This visual breakdown provides a clear overview of operational success distribution across historical launch sites.



Launch Success Rate for Highest Performing Site

Site-Specific Analysis

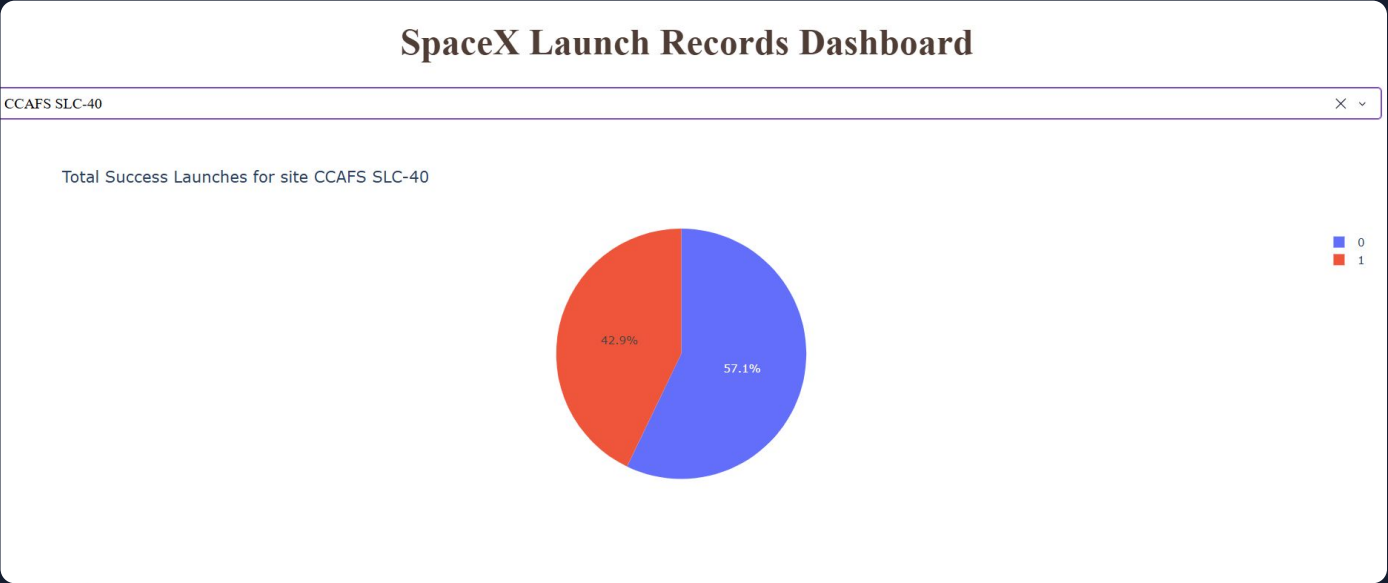
The interactive dashboard allows for site-specific performance analysis. By selecting individual launch facilities, the pie chart dynamically updates to reveal the exact ratio of successful (1) to failed (0) missions.

Top Performer: CCAFS SLC-40

CCAFS SLC-40 accounts for 42.9% of all successful landings.

Operational Excellence

This high success rate indicates highly optimized operational protocols at this location compared to other facilities.



Correlation Between Payload Mass and Mission Outcome

Visual Data Relationship

The interactive scatter plot visualizes the relationship between payload mass (kg) and mission outcome, using color-coded booster versions.

Booster Performance

Falcon 9 manages diverse weights effectively, while advanced booster classes like **B5** are optimized for heavy payloads.

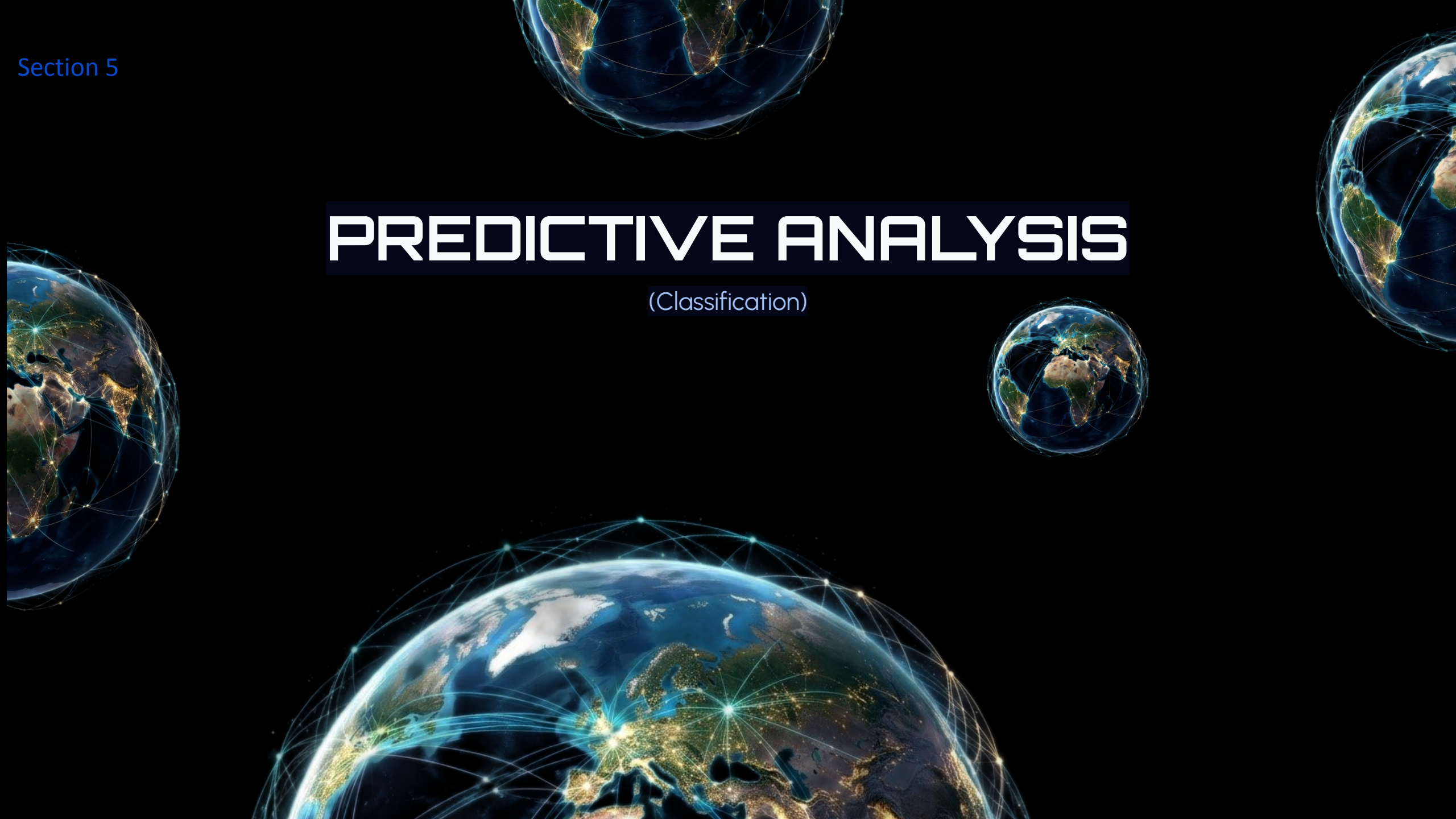
High Reliability

Maintains high reliability for the heaviest payload brackets, specifically those exceeding **8,000 kg**.



PREDICTIVE ANALYSIS

(Classification)



Classification Models Accuracy Comparison

Model Performance Analysis

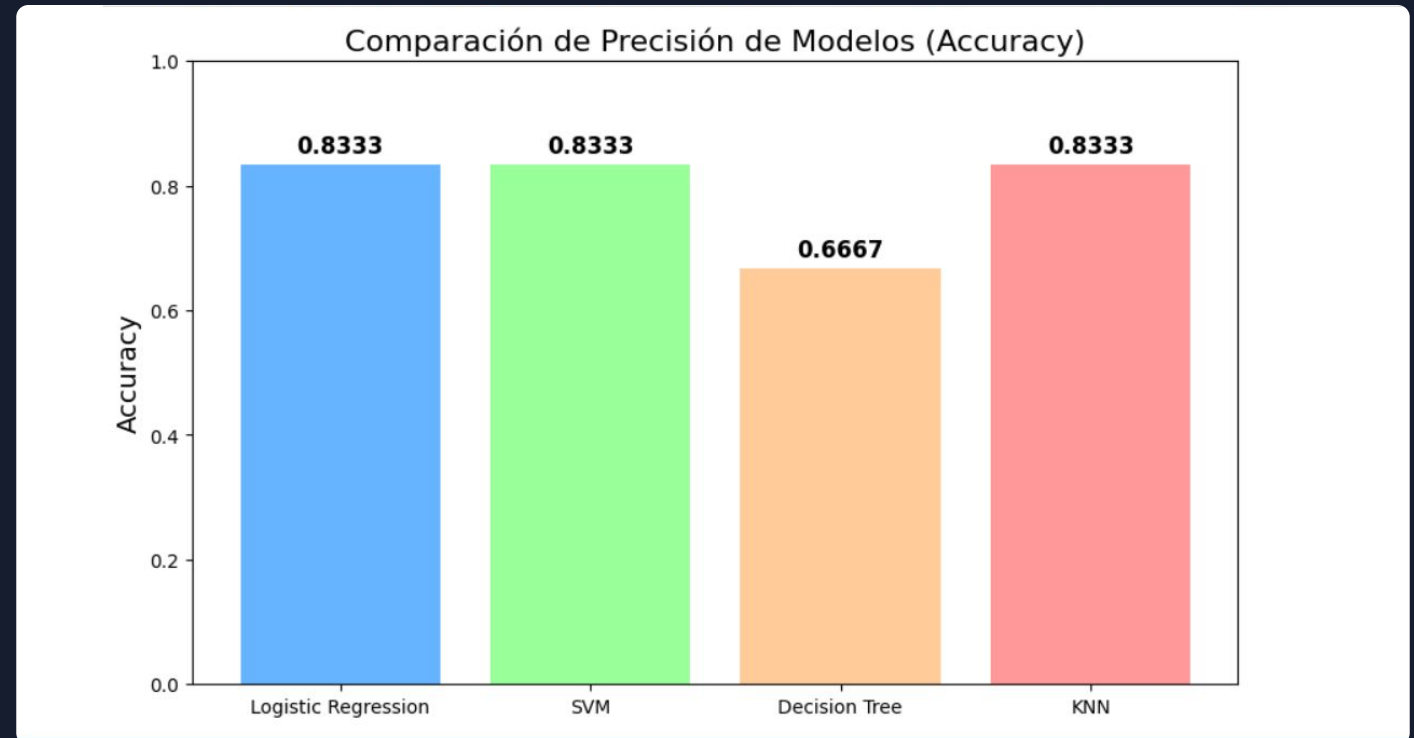
The bar chart visualizes the test set accuracy across four machine learning predictive models: Logistic Regression, SVM, Decision Tree, and KNN.

Top Performing Algorithms

Logistic Regression, SVM, and KNN performed equally well, achieving the peak classification accuracy of **83.3%** on unseen test data.

Least Effective Model

The Decision Tree model underperformed with **66.7%** accuracy, making it the least effective algorithm for this specific dataset.



Confusion Matrix: Best Performing Model (KNN)

Model Performance Peak

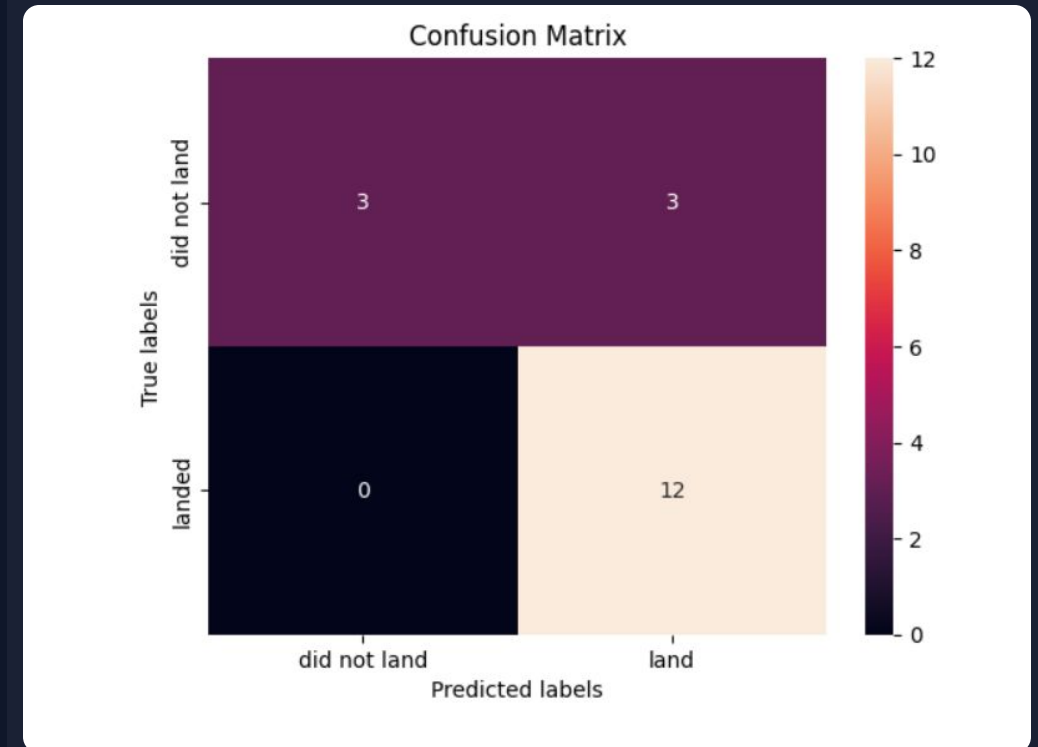
K-Nearest Neighbors (KNN) was selected to represent the best performing models, having achieved the peak test accuracy (**83.3%**) alongside Logistic Regression and SVM.

Perfect Positive Identification

The confusion matrix reveals that the model is perfectly accurate at identifying successful missions, successfully predicting all **12 positive landings** with zero False Negatives.

Predictive Error Analysis

The confusion matrix indicates **3 false positives** and no false negatives, suggesting strong sensitivity for identifying successful landings.



CONCLUSIONS

Technical Performance

- **Model Reliability:** KNN, Logistic Regression, and SVM reached 83.3% accuracy, effectively predicting outcomes.
- **Feature Impact:** Payload mass and booster version B5 are the strongest predictors of landing success.
- **Site Efficiency:** KSC LC-39A demonstrated the highest concentration of successful missions in the dataset.

Strategic Impact

- **Cost Optimization:** Predicting landing success allows for better estimation of the reusability price advantage.
- **Operational Maturity:** Significant improvement since 2013 highlights a rapid learning curve for Falcon 9.
- **Predictive Edge:** High True Positive rates prove that landing outcomes are now data-driven, not random.

Conclusion: The integration of ML and Geospatial Analysis highlights SpaceX's strong operational performance

APPENDIX: TECHNICAL ASSETS

Jupyter Notebooks (Analysis & Modeling)

- `Data_Collection_API.ipynb` & `Data_Collection_WebScraping.ipynb`: Raw data extraction and initial formatting.
- `EDA_Pandas_and_SQL.ipynb`: Exploratory Data Analysis utilizing Pandas and IBM Db2 SQL queries.
- `Interactive_Visualizations_Folium.ipynb`: Geospatial analysis generating interactive maps for launch site proximity.
- `Machine_Learning_Prediction.ipynb`: Training and evaluation of predictive classification models.

Python Scripts & Web Applications

- `spacex_dash_app.py`: Source code for the interactive Plotly Dash web application, featuring interactive dashboard callbacks.

Datasets Generated

- `dataset_part_1-3.csv`: Cleaned and pre-processed staging files.
- `spacex_launch_dash.csv`: Final dataset optimized for Dash application.

THANK YOU

FOR YOUR ATTENTION

- Lorena A. Benitez
- DATA SCIENCE PROFESSIONAL
- [linkedin.com/in/ninacocazero](https://www.linkedin.com/in/ninacocazero)
- github.com/NiNaCocaZero